AVN-101 hydrochloride

Cat. No.: HY-117046A

CAS No.: 1061354-48-0

Molecular Formula: $C_{21}H_{25}ClN_2$

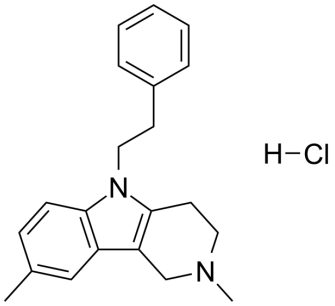
Molecular Weight: 340.89

Target: 5-HT Receptor; Histamine Receptor; Adrenergic Receptor

Pathway: GPCR/G Protein; Neuronal Signaling; Immunology/Inflammation

Storage: -20°C, sealed storage, away from moisture

* In solvent : -80°C, 6 months; -20°C, 1 month (sealed storage, away from moisture)



SOLVENT & SOLUBILITY

In Vitro

DMSO : 125 mg/mL (366.69 mM; Need ultrasonic)

Preparing Stock Solutions	Mass / Solvent / Concentration	1 mg	5 mg	10 mg
	1 mM	2.9335 mL	14.6675 mL	29.3350 mL
	5 mM	0.5867 mL	2.9335 mL	5.8670 mL
	10 mM	0.2933 mL	1.4667 mL	2.9335 mL

Please refer to the solubility information to select the appropriate solvent.

BIOLOGICAL ACTIVITY

Description

AVN-101 hydrochloride is a potent, brain-penetrant and orally active 5-HT7 receptor antagonist (K_i of 153 pM), with slightly lesser potency toward 5-HT6, 5-HT2A, and 5HT-2C receptors (K_i values of 2.04 nM, 1.56 nM, and 1.17 nM, respectively). AVN-101 hydrochloride also exhibits a rather high affinity toward histamine H1 (K_i of 0.58 nM) and adrenergic α2A, α2B, and α2C (K_i= 0.41-3.6 nM) receptors. AVN-101 hydrochloride can be studied in such diseases as general anxiety disorders, depression, schizophrenia, Alzheimer's disease, and multiple sclerosis[1].

IC_{50} & Target				
	5-HT_{2A} Receptor	5-HT_{2C} Receptor	5-HT_6 Receptor	5-HT_7 Receptor
	5-HT_7 Receptor 0.153 nM (Ki)	5-HT_{2C} Receptor 1.17 nM (Ki)	5-HT_{2A} Receptor 1.56 nM (Ki)	5-HT_6 Receptor 2.04 nM (Ki)
	5-HT_{2B} Receptor 10.6 nM (Ki)	5-HT_{1A} Receptor 61 nM (Ki)	5-HT_{1B} Receptor 720 nM (Ki)	H_1 Receptor 0.58 nM (Ki)
	Alpha-2B adrenergic receptor 0.41 nM (Ki)	Alpha-2A adrenergic receptor 1.77 nM (Ki)	Alpha-2C adrenergic receptor 3.55 nM (Ki)	Alpha-1B adrenergic receptor 9.4 nM (Ki)

	Alpha-1A adrenergic receptor 18.9 nM (Ki)	Alpha-1D adrenergic receptor 30.2 nM (Ki)

In Vitro

AVN-101 hydrochloride shows some ability to compete with radio-labeled ligands for both the histamine H1 (K_i = 0.58 nM) and H2 receptors (K_i = 89 nM) while being totally inactive on H3 receptor[1].

MCE has not independently confirmed the accuracy of these methods. They are for reference only.

In Vivo

AVN-101 (0.05 mg/kg; i.p.; once daily; for 4 days) hydrochloride reduces immobility in the forced swim test in mice[1].

AVN-101 (0.02-5 mg/kg; i.p.; once) hydrochloride prevents Scopolamine (HY-N0296)-induced amnesia in a passive avoidance test in mice when administered at a dose of 0.05 mg/kg[1].

AVN-101 (0.2, 1, and 5 mg/kg) hydrochloride increases the time spent in the open arms of the elevated plus maze in mice[1].

AVN-101 hydrochloride is poorly stable in murine plasma: by the end of 30-min incubation at room temperature, the residual amount of the original compound is reduced to 16%. At the same time, AVN-101 hydrochloride shows better stability in plasma samples taken from monkey and human: by the end of the 30-min incubation at room temperature, the residual amount of the original compound is reduced to only 45% and 68%, respectively[1].

MCE has not independently confirmed the accuracy of these methods. They are for reference only.

Animal Model:	Balb/C mice bearing porsolt forced swim and the tail suspension tests[1]
Dosage:	0.05 mg/kg
Administration:	i.p.; once daily; for 4 days
Result:	Reduced immobility in the forced swim test in mice.

Animal Model:	Male BALB/c mice were used with amnesiac models based on either Scopolamine-induced memory impairment[1].
Dosage:	0.02 mg/kg, 0.05 mg/kg, 0.2 mg/kg, 0.5 mg/kg, 1 mg/kg, and 5 mg/kg
Administration:	i.p.; once
Result:	Prevented scopolamine-induced amnesia in a passive avoidance test in mice.

REFERENCES

[1]. Alexandre V Ivachtchenko, et al. AVN-101: A Multi-Target Drug Candidate for the Treatment of CNS Disorders. J Alzheimers Dis. 2016 May 25;53(2):583-620.

Caution: Product has not been fully validated for medical applications. For research use only.

Tel: 609-228-6898 Fax: 609-228-5909 E-mail: tech@MedChemExpress.com

Address: 1 Deer Park Dr, Suite F, Monmouth Junction, NJ 08852, USA